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Taming semantic segmentation for effective transferable attack against object detection

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ABSTRACT

Previous transfer-based black-box attacks have primarily focused on image classification tasks, making limited progress in the domain of object detection, where attack performance generally remains constrained. To address this challenge, we propose a Cross-Task Synergistic Transferable Attack method, which enhances attack performance against black-box object detection models from a multi-task perspective. Our approach begins by selecting semantic segmentation, a task closely related to object detection, as an auxiliary task. We then design the Cross-Task Semantic Representation Alignment module to perform fine-grained distillation of adversarial information, supplementing the insufficient information available from the object detection task. During the attack process, we introduce the Iterative Cross-Model Gradient Averaging strategy. This module consists of Cross-Model Gradient Collection and Cross-Model Gradient Averaging components, which enhance the stability of the attack process through multi-step cross-model gradient aggregation and gradient conflict optimization. Extensive experiments demonstrate that, compared with state-of-the-art methods, our approach achieves the highest attack success rate on object detection datasets such as COCO and VOC, showcasing superior attack effectiveness.

1. Introduction

Object detection is a cornerstone task in computer vision, dedicated to recognizing and pinpointing objects within images. It underpins a myriad of applications that are integral to our daily lives, spanning from security surveillance and autonomous driving to medical diagnosis (Nguyen et al., 2025). While state-of-the-art object detection technologies are rooted in deep learning frameworks, these intricate neural networks are susceptible to meticulously crafted adversarial perturbations. Therefore, it is crucial to study the robustness of object detection techniques to enhance our understanding of this critical aspect and thereby safeguard the safe and reliable application of object detection models (Wang et al., 2024).

Although adversarial attacks on classification tasks have made significant progress, the architectural and procedural differences between object detection frameworks and classification models (such as the distinction between one-stage and two-stage detectors) make it difficult to directly apply existing methods to object detection. The inherent complexity of object detection further increases the difficulty of generating effective adversarial examples. In recent years, transferability has become a commonly used strategy for attacking object detection models, where adversarial examples generated on one model can successfully attack other models. However, the performance of existing transfer-based attack methods heavily depends on the degree of architectural similarity between the surrogate model and the target model: when the architectures are the same, the attack performance is strong, but when the architectures differ significantly, the transfer performance drops notably. This makes the attack performance highly reliant on the quality of the surrogate model and the degree of architectural alignment, thereby limiting its overall effectiveness in cross-architecture transfer scenarios.

Instead of struggling to find the most similar surrogate model, this paper proposes leveraging related auxiliary tasks to compensate for the inherent adversarial information deficiency of the target model, thereby enhancing the transferability of adversarial examples. Inspired by multi-task learning (Liu et al., 2019), when tasks are somewhat related, multi-task learning can utilize cross-domain transfer learning mechanisms during training to transfer knowledge learned from one task to another. This helps to compensate for the lack of specific domain knowledge and significantly enhances the robustness and generalization of feature extraction. We apply this finding to adversarial attack tasks by preparing white-box surrogate models that learn more comprehensive adversarial

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information from multiple tasks, thereby enhancing the transferability of attacks. However, to adopt this approach, two key issues need to be addressed. (1) Extracting effective adversarial information from related tasks: Methods need to be designed to obtain valuable adversarial information from auxiliary tasks related to the target task, thereby compensating for the adversarial information deficiency in the target model; (2) Optimizing conflicting adversarial gradients among multiple surrogate models: In a multi-model setup, adversarial gradients may conflict, so strategies are needed to coordinate and optimize these gradients to ensure consistency across different models.

To achieve this, we propose a new Cross-Task Synergistic Transferable Attack (CTSTA) framework to generate transferable adversarial examples. The main idea is to extract and combine adversarial information from related auxiliary tasks to improve the effectiveness and transferability of adversarial attacks. Our approach involves supplementing the target task's adversarial knowledge with cross-task knowledge distillation from related tasks and designing a strategy to optimize and integrate adversarial gradient information from multiple surrogate models.

More specifically, to address the first issue, the most intuitive idea is to use the framework of knowledge distillation, but traditional knowledge distillation methods typically involve a teacher model and a student model within the same task domain, with the primary goal of enhancing the performance of the student model. Our objective is different: we aim to use related auxiliary tasks to compensate for the deficiencies in adversarial information inherent in the target model. To achieve knowledge distillation, many methods (Qian et al., 2024) attempt to manually select feature links between the teacher and student. Jang et al. (2019) proposed the Learning Transfer meta-network, which optimizes the distillation process by automatically identifying feature links, but the training is expensive. Thus, we propose a novel Cross-Task Semantic Representation Alignment (CTSRA) module, where the auxiliary model acts as the teacher model and the target model serves as the student model. This method assesses feature similarity through Semantic Consistency Measurement and uses Semantic Consistency-Driven Knowledge Transfer to dynamically determine the optimal distillation strength between different teacher-student feature pairs. This facilitates the effective transfer of valuable adversarial knowledge from the teacher model to the student model, enabling the target model to learn more comprehensive adversarial information during the attack process.

To address the second issue, we aim for a comprehensive gradient optimization strategy. Research has shown that ensemble-based transfer attacks can significantly improve the transferability of adversarial examples (Chen et al., 2023). However, due to differences in gradient space between different surrogate models, directly averaging these gradients may lead to a reduction in the effectiveness of adversarial perturbations and limit the transferability of adversarial examples. To address this issue, we propose an Iterative Cross-Model Gradient Averaging (ICGA) strategy. The ICGA framework comprises two key components: Cross-Model Gradient Collection and Cross-Model Gradient Averaging. The Cross-Model Gradient Collection module alternately gathers multi-step gradient information from different surrogate models to explore a more comprehensive and generalized gradient space. The Cross-Model Gradient Averaging module integrates cross-model multi-step gradients while performing projection processing on conflicting gradients, thereby significantly enhancing the stability of the attack process.

The main contributions of our work can be summarized as follows:

- To address the issues of insufficient adversarial information and poor transferability in surrogate models, we design a Cross-Task Semantic Representation Alignment that leverages auxiliary tasks to supplement information, enhancing the generalization and transferability of the generated adversarial examples.
- During the attack phase, to tackle gradient conflicts and instability across multiple models, we propose an Iterable Cross-Model Gradient Averaging strategy that alternately collects and fuses multi-step gradients to improve gradient compatibility and attack stability.
- We conduct extensive and detailed experiments, demonstrating that our method achieves superior attack performance over several datasets.

2. Related work

2.1. Black-box adversarial attacks

Adversarial attack methods are mainly divided into two categories: white-box attacks and black-box attacks. White-box attacks assume that the attacker has full access to the internal information and parameters of the target model (Keyizhi et al., 2025; Wang et al., 2025). In contrast, black-box attacks are conducted solely through interaction with the model's outputs (Li et al., 2025; Qian et al., 2024). Since attackers in real-world scenarios often lack detailed internal information about the target model, black-box attacks have attracted increasing attention. Black-box attacks can be mainly categorized into query-based attacks and transfer-based attacks. Query-based attacks involve sending specific query samples to the target model, analyzing its outputs, and inferring its internal operations and potential vulnerabilities. These attacks typically require a large number (hundreds or thousands) of queries, which can be resource-intensive and pose potential detection risks. Transfer-based attacks leverage existing data or models to transfer information onto the target model, infer its behavior, and launch attacks. In our work, we primarily focus on transfer-based attacks.

To alleviate the overfitting issue of traditional white-box attack methods and enhance the transferability of adversarial examples, FGSM (Goodfellow et al., 2014) proposes the fast gradient sign method for quick and simple generation of adversarial examples. PGD (Madry et al., 2017) advances projected gradient descent as a robust first-order adversary leveraging local first-order information from networks. CW (Carlini & Wagner, 2017) introduces three new attack methods based on L_0, L_2, and L_∞ norms to successfully attack distilled networks. MIM (Dong et al., 2018) based on momentum iteration generation adversarial sample method to stabilize the updating process. NIM (Lin et al., 2019) incorporates Nesterov accelerated gradient into iterative attacks. VIM (Wang & He, 2021) uses variance regularization to adjust gradients, enhancing the transferability of adversarial attacks by modifying gradients. In addition to gradient modification, some methods enhance the transferability of adversarial examples by augmenting input images, such as DIM (Xie et al., 2019), which applies diversified input transformations to design enhanced adversarial attack strategies across different networks. PIM (Gao et al., 2020) employs a patch-based iterative algorithm, adjusting step sizes with amplification factors and using a projection kernel to distribute gradient overflow to surrounding regions. TIM (Dong et al., 2019) utilizes translation-invariant attack methods to generate more transferable adversarial examples, while SIM (Lin et al., 2019) designs scale-invariant attack methods. However, in cases where there are significant differences in model architectures, the effectiveness of some methods may not be ideal. While previous studies have primarily focused on adversarial attacks targeting image classifiers, our work shifts the focus to exploring the security of object detection models, given their wide usage and critical importance.

2.2. Attacks against object detectors

Object detectors are inherently more complex and face greater challenges compared to image classification tasks. Due to the diverse architectures and complex processing pipelines in object detection, existing adversarial attack methods designed for image classification tasks are difficult to apply directly to object detection. Existing attack methods around the object detection model, such as Dense Adversarial Generation (DAG) (Xie et al., 2017), propose optimizing loss functions to generate adversarial perturbations on pixel/proposal sets. RAP (Li et al., 2018) introduces a loss function combining label loss and shape loss, optimizing attack images through gradient-based iterative methods. CAP

(Zhang et al., 2020) devises contextual adversarial perturbations using contextual background loss to generate adversarial samples without relying on true labels. However, these methods are tailored for region proposal networks, limiting their applicability to proposal-based (two-stage) object detectors, depending on the detector type used. In contrast, LiDAttack (Chen et al., 2025) precisely controls perturbation points through genetic algorithms combined with simulated annealing. Pi et al. (2025) improve attack success rate by optimizing the starting point through warm-up attacks and employing a decision boundary enhancement strategy. RLAB (Sarkar et al., 2025) uses a dual-action mechanism of a reinforcement learning agent to simultaneously identify sensitive regions for adding distortions and remove ineffective noise. Ennaji et al. (2026) adaptively identify sensitive features through change-point detection and causal analysis. However, these attack methods often overfit the specific white-box models used and may not achieve better attack performance on other black-box models. We propose a Cross-Task Synergistic Transferable Attack method to address the deficiencies in adversarial information of the target model, thereby enhancing its generalization ability. Additionally, we use a gradient optimization strategy to refine perturbations, effectively improving the success rate of adversarial attacks on black-box models.

2.3. Multi-task learning

Multi-task learning (MTL) is a widely used machine learning approach, whose core advantage lies in learning shared representations by jointly training related tasks. Our focus is on leveraging these shared representations to transfer knowledge learned from other tasks to a target task, thereby improving the generalization of single tasks. (Zhang & Yang, 2021) and Vandenhende et al. (2021) have shown that, compared with single-task learning, multi-task training allows related tasks to share feature representations and benefit from each other, leading to improved performance. Further, the work of Mao et al. (2020) demonstrates that multi-task training enables models to learn more robust features, indicating that cross-task shared representations not only enhance single-task performance but also improve the model's resilience to noise or adversarial perturbations. In addition, Hu et al. (2017) research shows that models trained on attribute recognition tasks can learn stable features that enhance the performance of face recognition tasks. When tasks exhibit certain correlations, multi-task learning models can further strengthen robustness, resulting in more resilient adversarial features. The emergence of foundation segmentation models further expands the potential of this cross-task paradigm. CCNet (Huang et al., 2019) efficiently captures global context through criss-cross attention; CTNet (Li et al., 2021) achieves effective integration via spatial and channel contextual interactions with adaptive fusion; VRP-SAM (Sun et al., 2024b) enables precise segmentation with minimal parameters through a visual reference prompt encoder. Specific studies (Chen et al., 2022; Lv et al., 2022) highlight significant correlations between tasks such as object detection and semantic segmentation, demonstrating that MTL can notably improve the performance of mutually related tasks like object detection and semantic segmentation.

Existing researches primarily focus on white-box attacks targeting MTL models across multiple tasks. For instance, methods like MTA (Guo et al., 2020) utilize generators to produce universal adversarial perturbations that can attack multiple tasks simultaneously. ADGS (Liu et al., 2023) enhances the generalization of adversarial samples across different tasks by applying attention dispersion and gradient sharpening. DGBA (Zhang et al., 2023) proposes dynamic gradient balancing attack methods to maximize the potential of adversarial samples in white-box attacks on MTL models. Different from approaches that aim to attack multiple tasks simultaneously, some studies explore adversarial learning from the perspective of cross-task transfer. For example, Marvin (Klingner et al., 2020) introduces a self-supervised monocular depth estimation auxiliary task during training and leverages multi-task learning to improve the robustness of semantic segmentation models against noise and adversarial attacks; (Kong et al., 2024) enhances the transferability of adversarial examples in black-box scenarios by sharing gradient information across tasks and constructing a cross-task gradient mapping mechanism. Inspired by these works, we do not aim to attack multiple tasks at the same time. Instead, we exploit the cross-task knowledge sharing and transfer enabled by multi-task learning to more effectively focus on attacking a single object detection task, thereby significantly improving the transferability of adversarial examples in object detection scenarios.

The core idea of gradient-based meta-learning is to perform meta-training across multiple tasks, enabling the model to learn how to adapt quickly and efficiently. MAML (Finn et al., 2017) uses a two-level gradient optimization to learn initial parameters that allow fast adaptation to new tasks. Reptile (Nichol et al., 2018) approximates the optimal initialization by averaging the directions of multiple SGD updates, offering lower computational cost. MT-Net (Baik et al., 2021) enhances the multi-task learning ability by introducing task-specific parameter masks into the MAML framework. Integrating gradient-based meta-learning into multi-task learning helps improve the adaptability and optimization efficiency of shared representations across tasks.

3. Methodology

3.1. Framework overview

In this section, we propose a task-driven and novel Cross-Task Synergistic Transferable Attack (CTSTA) method for object detection, which contains a Cross-Task Semantic Representation Alignment module (CTSRA) and an Iterable Cross-Model Gradient Averaging (ICGA) module. Firstly, in Section 3.2, we describe the task definition. In Section 3.3, we describe the attack setting adopted in this study. In Section 3.4, we introduce a CTSRA module to distill relevant information from auxiliary tasks, constructing an object detection surrogate model with enhanced performance. To further enhance the attack effectiveness and transferability of adversarial samples, we integrated multiple information-enhanced surrogate models. However, during the integration process, gradient conflicts between different models became a challenge. Thus, in Section 3.5, we present an ICGA algorithm, including Cross-Model Gradient Collection and Cross-Model Gradient Averaging strategies to stabilize the update process and improve the transferability of adversarial samples.

3.2. Task definition

In object detection, the ground truth annotation results of an image x are typically represented in the following form:

$$\mathcal{F}(x) = \left\{ B_i(x), y_i(x) \right\}, i = 1, 2, \ldots, K. \tag{1}$$

where $B_i(x) = \left\{ B_i^{x_1}, B_i^{y_1}, B_i^{x_2}, B_i^{y_2} \right\}$ represents the position coordinates of the detection box, where (x_1, y_1) denotes the coordinates of the top-left corner and (x_2, y_2) denotes the coordinates of the bottom-right corner. The variable $y_i(x)$ takes values from 1 to C, indicating the category information of the object, with C representing the total number of possible classes in the dataset. K denotes the total number of detection boxes present in the image x.

The detection results of image x from the object detector can be represented as:

$$\mathcal{F}'(x) = \left\{ B_i'(x), z_i'(x) \right\}, i = 1, 2, \ldots, K. \tag{2}$$

where $z_i'(x) = \left\{ z_i'^1, z_i'^2, \ldots, z_i'^C \right\}$ is an output vector containing probabilities for each category. After undergoing softmax processing, the predicted category is selected as the one with the highest probability:

$$y_i'(x) = \underset{i}{argmax}\left(softmax\left(z' \right)_i \right). \tag{3}$$

3.3. Attack strategy

In this section, we describe the attack setting and explain the corresponding loss function design and perturbation update methods.

In the targeted attack setting, we detect K objects in the original image x and randomly select one, such as the kth object, as the attack target. We then change its category to the target class according to the following selection rules:

$$y'_{tgt} = \{c \mid c \in C_{\text{total}}, c \notin C_{\text{class}}\}. \tag{4}$$

where Ctotal is the set of all categories in the dataset, and Cclass is the set of categories present in the current image. y'_{tgt} denotes the set of target categories. This setup helps eliminate the possibility of mistakenly counting existing objects as successful attacks. The modified result is taken as the detection truth value of image x, denoted as B_{gt}.To calculate the loss and generate the adversarial sample, we define the following objective functions. Specifically, the loss is formulated as:

$$L_\theta(x) = L\big(B_{gt}, \theta'(x+\xi)\big). \tag{5}$$

based on which the adversarial sample x_{adv} is generated by solving the following optimization problem:

$$x_{adv} = \underset{\|\xi\|_p<\epsilon}{\arg\min}\, L\big(B_{gt}, \theta'(x+\xi)\big). \tag{6}$$

where ξ denotes the adversarial perturbation added to the input x, and ϵ is a constant constraining the l_p-norm of ξ , with p set to ∞ in this paper. Here, $L\big(B_{gt}, \theta'(x+\xi)\big)$ represents the training loss corresponding to the perturbed input $x+\xi$ and the ground truth label B_{gt}.

3.4. Cross-task semantic representation alignment

In this section, we propose a Cross-Task Semantic Representation Alignment (CTSRA) module from a multi-task learning perspective. First, at the pixel-level fine-grained spatial semantic level, the dense supervision provided by the segmentation task effectively compensates for the spatial granularity limitation of object-level annotations in object detection, enabling the generated perturbations to precisely target key regions of the object. Second, at the level of universal semantic representation, the feature alignment mechanism shifts the focus toward modeling the global semantic structure of the target object, filtering out architecture-specific noise and encouraging perturbations to concentrate on more stable and generalizable semantic features, thereby alleviating the overfitting issue commonly observed in white-box surrogate models.

We regard the semantic segmentation model as the teacher model and the target detection model as the student model. The feature set extracted by the teacher model in each stage of the convolutional neural network is represented by $f^T = \{f_1^T, f_2^T, \ldots, f_M^T\}$, where M represents the number of teacher features. The feature set extracted by the student model in each stage of the convolutional neural network is represented by $f^S = \{f_1^S, f_2^S, \ldots, f_N^S\}$, where N represents the number of student features. Each feature has its own feature size and channel dimension, represented by $\mathbb{R}^{H\times W\times d}$, where H, W, and d represent the height, width, and channel dimension corresponding to the feature, respectively. The goal of our method is to identify the semantic consistency between all possible pairs of features in the teacher feature set and the student feature set, a total of $M \times N$ pairs, and transfer the valid knowledge from the teacher model to the student model based on the identified semantic consistency.

As shown in Fig. 1, CTSRA includes two components: Semantic Consistency Measurement (SCM) and Semantic Consistency-Driven Knowledge Transfer (SCDKT). The SCM strategy first performs global average pooling on the features, calculates the overall representation of each feature, and measures the semantic consistency of feature pairs between the f^T and f^S two feature sets. Next, the SCDKT strategy applies channel-wise average pooling to reduce the dimensionality of the feature maps, extract the main features, and compute the channel-level feature distance. This, combined with the semantic consistency obtained from the SCM module, guides the strength of knowledge transfer. The overall process of the CTSRA algorithm is shown in Algorithm 1.

Algorithm 1 Cross-task semantic representation alignment training procedure.

Require:

Input: pre-trained object detection model S, pre-trained semantic segmentation teacher model T, L_{CTSRA} weight α, learning rate τ, epoch C.

Output: Updated parameters of the object detection model θ_S.

1: **for** c **in** $1, \ldots, C$ **do**
2: Extract teacher feature set f^T and student feature set f^S.
3: Calculate query q and key k based on Eq. (7) and Eq. (8).
4: Calculate attention score η according to Eq. (9).
5: Calculate L_{CTSRA} according to Eq. (12).
6: Obtain the total training loss $L_{Student}$ according to Eq. (13).
7: Update the parameters θ_S^c of the object detection model according to Eq. (14).
8: **end for**
9: **return** θ_S

a) ***Semantic Consistency Measurement***. Each teacher feature is represented as a query q_m, while each student feature is represented as a key k_n. The mathematical expressions for the query q_m and key k_n are as follows:

$$q_m = g\big(U_m \cdot \psi\big(f_m^T\big)\big). \tag{7}$$

$$k_n = h\big(V_n \cdot \psi\big(f_n^S\big)\big). \tag{8}$$

In the query-key mechanism, we use the global average pooling function ψ to average over the spatial dimensions, compressing the spatial features into a global semantic representation. Functions $g(\cdot)$ and $h(\cdot)$ denote the activation functions for the query and key, respectively. $U_m \in \mathbb{R}^{d\times d_m^T}$ represents the linear transformation parameters for the query, where d_m^T denotes the channel dimension of the teacher feature, and d represents the output feature channel dimension after transformation. Similarly, $V_n \in \mathbb{R}^{d\times d_n^V}$ represents the linear transformation parameters for the key, where d_n^V denotes the channel dimension of the student feature.

Because teacher and student features typically exhibit different properties across different levels. For instance, low-level visual features may capture simple lines or edges in an image, while high-level visual features may represent more complex objects or scenes. Therefore, to better capture and utilize information across these different levels, we apply distinct linear transition weights to each feature. Specifically, we first compute a holistic representation of each feature using global average pooling. Based on these pooled representations, we measure the semantic consistency between feature pairs via a query-key mechanism, where the matching score $\text{Score}_{m,n}$ between the mth teacher feature and the nth student feature is defined as:

$$\text{Score}_{m,n} = \frac{q_m^{\top} O_m k_{m,n} + (l_m^{\mathrm{T}})^{\top} l_n^{\mathrm{S}}}{\sqrt{d}}. \tag{9}$$

In this formulation, $O_m \in \mathbb{R}^{d\times d}$ is a bilinear weight matrix that maps and integrates query and key vectors from different feature spaces to capture correlated information. Additionally, l_m^{T} and l_n^{S} denote the position encodings for the teacher and student features, respectively, sharing common information across levels. We divide the dot product by the square root of the dimension $\sqrt{d}$ to prevent excessively large values, which could otherwise trigger the vanishing gradient problem during backpropagation.

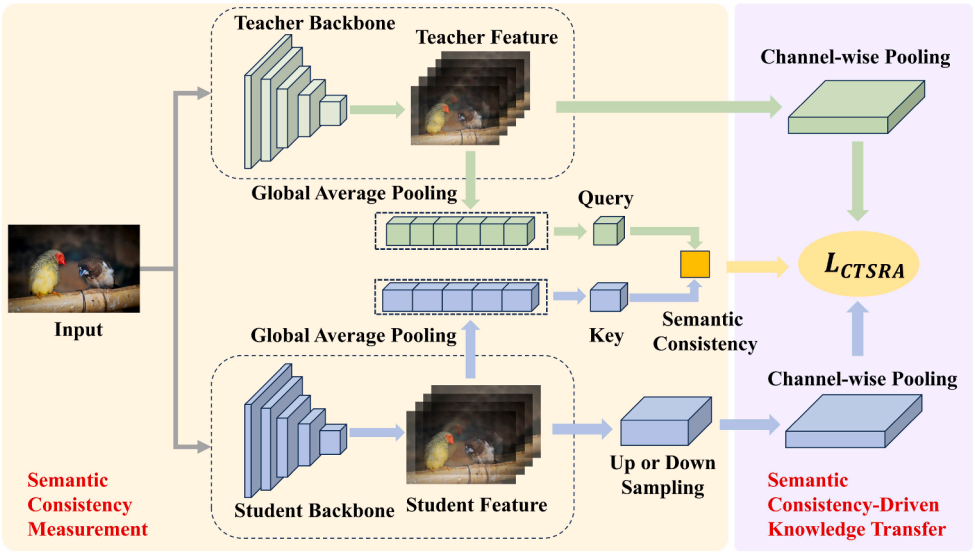


Fig. 1. Proposed cross-task semantic representation alignment network framework. Among them, the teacher model is the auxiliary model, namely the semantic segmentation model, and the student model is the object detection model. The Semantic Consistency between each teacher-student feature pair is calculated through the Semantic Consistency Measurement module, and the intensity of knowledge distillation is known in the Semantic Consistency-Driven Knowledge Transfer module.

To transform the raw matching scores into probability weights $\eta_{m,n}$, thereby dynamically filtering the most relevant feature levels to guide precise knowledge transfer, we apply Softmax normalization:

$$\eta_{m,n} = \frac{\exp(\text{Score}_{m,n})}{\sum_{j=1}^{N} \exp(\text{Score}_{m,j})}. \tag{10}$$

which quantifies the semantic correlation strength of the mth teacher feature relative to each student feature level. To facilitate the subsequent knowledge distillation process, these weights are further assembled into the vector η_m:

$$\eta_m = [\eta_{m,1}, \eta_{m,2}, \ldots, \eta_{m,N}]. \tag{11}$$

Through this process, the semantic consistency η_m between the mth teacher feature and all student features is fully established, serving as a robust weighting basis for cross-task knowledge transfer.

b) ***Semantic Consistency-Driven Knowledge Transfer***. Based on the computed η_m, the teacher model can selectively distill relevant knowledge to the student model. The final formula for the CTSRA module is as follows:

$$L_{CTSRA} = \sum_{m} \sum_{n} \eta_{m,n} \cdot \left\| \varphi\left(f_m^T\right) - \varphi\left(\tilde{f}_n^S\right) \right\|_2. \tag{12}$$

where $\varphi(\cdot)$ represents a combined function of channel-wise average pooling and L_2 normalization, used to process teacher and student features.

The function $\varphi(\cdot)$ represents channel-wise average pooling followed by feature fusion to highlight spatial structure information, which assists in computing feature distances at the channel level. Additionally, $\tilde{f}_n^S$ is sampled or downsampled from student features f_n^S, aiming to align the spatial dimensions of teacher and student features for computing channel-level feature distances. $\eta_{m,n}$ denotes the semantic consistency weight between the mth teacher feature and the nth student feature. Through the weighted contribution of $\eta_{m,n}$, discriminative knowledge at key regions is more effectively and selectively transferred to the student model, thereby enhancing the distillation performance.

By adding the CTSRA distillation item loss to the training loss function of the student model, the final total loss function can be expressed as:

$$L_{Student} = L_{Det} + \alpha \cdot L_{CTSRA}. \tag{13}$$

where L_{Det} represents the classification and regression loss of the target detection model, and α controls the weight parameter of the proposed distillation loss on the total loss. During training, both the student network and the attention-based network are guided by this total loss to jointly enhance the model's performance.The parameters of the student model are denoted as θ_s, and the update process is represented as:

$$\theta_s^c = \theta_s^{c-1} - \tau \cdot \nabla_{\theta_s^{c-1}} \left\{ L_{Student}\left(\theta_s^{c-1}\right) \right\}. \tag{14}$$

θ_s^c denotes the student model parameters after the cth iteration update, θ_s^{c-1} represents the parameters from the previous iteration, τ is the learning rate.

3.5. Iterable cross-model gradient averaging

In the previous section, we enhanced the adversarial information of object detection surrogate models through the CTSRA strategy. To further improve the attack effectiveness and transferability of the generated adversarial samples, we aim to integrate multiple such enhanced models. However, the complementary information provided by multiple models helps to improve the generalization ability of the perturbations, but this also requires reasonable coordination of the differences between the gradients of each model to prevent information cancellation. The gradient spaces and directions of different models have their own characteristics and discrepancies, which can cause gradient interference or even oscillations during optimization. Such gradient inconsistency makes it difficult for the perturbations to stably converge to regions with better attack effectiveness. To address these issues, we propose the Iterable Cross-Model Gradient Averaging (ICGA) strategy. Unlike parallel averaging of gradients from multiple models at the same perturbation point, ICGA updates the perturbation in an alternating manner, such that the gradient of each subsequent model is re-evaluated at a perturbation point that has already been validated as effective by the preceding model. In this process, gradient components that strongly conflict with the existing update direction are naturally suppressed, while directions that simultaneously increase the attack loss across multiple models are preserved and progressively reinforced. As a result, the perturbation is continuously guided toward shared adversarial directions that are jointly sensitive across models, automatically filtering out model-specific conflicting gradients and reducing gradient cancellation and oscillations. Meanwhile, it applies projection to conflicting gradients, ensuring stability and consistency in the gradient update direction. In

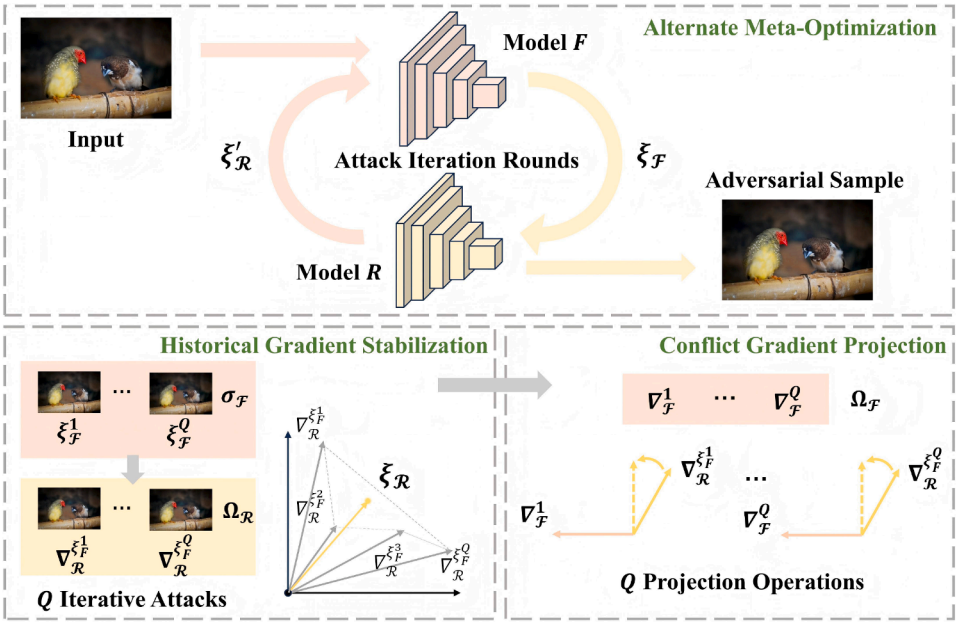


Fig. 2. The proposed ICGA algorithm framework: a) Cross-Model Gradient Collection: alternately obtain multi-step gradient information from different surrogate models; b) Cross-Model Gradient Averaging: integrate multi-step gradients across models while performing projection processing on conflicting gradients.

this section, we integrated two enhanced surrogate models, represented by $\mathcal{F}$ and $\mathcal{R}$ respectively, $U = \{\mathcal{F}, \mathcal{R}\}$ represents the set of agent models.

ICGA consists of two key modules: Cross-Model Gradient Collection and Cross-Model Gradient Averaging. The Cross-Model Gradient Collection module obtains gradients from multiple surrogate models to ensure diversity in sampling and complementary perspectives across models. The Cross-Model Gradient Fusion module fuses these collected gradients through averaging to reduce interference caused by conflicting directions, thereby enhancing the stability and consistency of the attack process. An illustration of the ICGA strategy is shown in Fig. 2.

a) ***Cross-Model Gradient Collection***. This module aims to enhance the compatibility of gradients across different models by alternately collecting adversarial gradient information from multiple surrogate models, thereby constructing adversarial perturbations with stronger transferability. Inspired by the "alternating update" concept in meta-learning, in each iteration, adversarial gradients are alternately collected from one model to update the perturbation, which is then transferred to the other model for further optimization.

Let $L_{\mathcal{F}}$ denote the adversarial loss function for model $\mathcal{F}$, and $L_{\mathcal{R}}$ denote the adversarial loss function for model $\mathcal{R}$. At the beginning of the attack, a model is randomly selected from the set U, the attack alternates in each iteration, and the corresponding gradient of the adversarial loss function is calculated to update the adversarial disturbance. Select model $\mathcal{F}$ first as an example, the update formula for the perturbation is as follows:

$$\xi_{\mathcal{F}} = \Pi_{\epsilon}\left\{\xi - \mu \cdot sign\left(v_1 \cdot L_{\mathcal{F}}\left(f'_{tgt}(x), f'(x+\xi)\right)\right)\right\}. \tag{15}$$

where ξ is the initialized perturbation, v_1 is the update hyperparameter, μ is the update step size, ϵ indicates the perturbation threshold, $\Pi_{\epsilon}(\cdot)$ represents the projection function under the L_{∞} constraint. Based on the aforementioned attack, the perturbation $\xi_{\mathcal{F}}$ for model $\mathcal{F}$ is computed. This perturbation is then transferred to another model $\mathcal{R}$, upon which $\mathcal{R}$'s adversarial loss is computed and the perturbation is further updated:

$$\xi_{\mathcal{R}} = \Pi_{\epsilon}\left\{\xi_{\mathcal{F}} - \mu \cdot sign\left(v_2 \cdot L_{\mathcal{R}}\left(f'_{tgt}(x), f'\left(x' + \xi_{\mathcal{F}}\right)\right)\right)\right\}. \tag{16}$$

where $x' = x + \xi$, v_2 is the update hyperparameter, $\xi_{\mathcal{R}}$ is the adversarial perturbation obtained in this iteration. By alternately collecting gradients from different surrogate models and optimizing the perturbation, better gradient compatibility is achieved.

During the process of updating the adversarial perturbation, the gradient directions from different surrogate models may conflict, causing oscillations in the update process and making it difficult to converge to an optimal solution. To alleviate the instability in the perturbation update direction, we further collect the historical perturbation trajectories. In each iteration of the optimization process, define an internal iteration count Q for the selected model branch, such as $\mathcal{F}$. Subsequently, compute adversarial perturbations Q times consecutively according to Eq. (15):

$$\sigma_{\mathcal{F}} = \left\{\xi_{\mathcal{F}}^1, \xi_{\mathcal{F}}^2, \ldots, \xi_{\mathcal{F}}^Q\right\}. \tag{17}$$

Transfer the Q adversarial perturbations computed to the model branch $\mathcal{R}$, add them to the image x to generate adversarial samples, and subsequently compute the corresponding Q gradients:

$$\Omega_{\mathcal{R}} = \left\{\nabla_{\mathcal{R}}^{\xi_F^1}, \nabla_{\mathcal{R}}^{\xi_F^2}, \ldots, \nabla_{\mathcal{R}}^{\xi_F^Q}\right\}. \tag{18}$$

b) ***Cross-Model Gradient Fusion***. During the process of collecting gradients across multiple models, significant directional conflicts often arise among the gathered gradients, which can lead to oscillations or even failure in the perturbation optimization process. The Cross-Model Gradient Fusion module is designed to effectively address such gradient conflicts by integrating current and historical gradients and employing a projection alignment strategy, thereby enhancing the stability and convergence of perturbation updates.

We first define the following perturbation update formula to update the perturbation on the other model branch $\mathcal{R}$, where the current gradient is fused with historical gradients in a weighted manner:

$$\xi_{\mathcal{R}} = \Pi_{\epsilon}\left\{\xi_{\mathcal{F}}^Q - \mu \cdot sign\left[v_2 \cdot \left(\nabla_{\mathcal{R}}^{\xi_F^Q} + v_3 \cdot \sum_{i=1}^{Q-1} \nabla_{\mathcal{R}}^{\xi_F^i}\right)\right]\right\}. \tag{19}$$

The hyperparameter v_3 is used to balance the weights of these gradients. Since historical adversarial gradients provide supplementary gradient information without dominating the primary update direction, v_3 is set to a small value.

In multi-task learning, gradient conflict is a major optimization challenge: gradients from different tasks may point in opposing directions,

which can severely degrade performance. Similarly, when generating adversarial examples using multiple surrogate models, a parallel issue of gradient conflict arises. To address this, we first assess whether a conflict exists by computing the cosine similarity between gradients. If a conflict is detected, we project the gradients to reduce interference and effectively align them. To implement this, we record the gradients of the Q adversarial perturbations from model $\mathcal{F}$ as a set:

$$\Omega_{\mathcal{F}} = \left\{ \nabla_{\mathcal{F}}^{1}, \nabla_{\mathcal{F}}^{2}, \ldots, \nabla_{\mathcal{F}}^{Q} \right\}. \tag{20}$$

Based on this set, we apply the gradient projection technique to update $\Omega_{\mathcal{R}}$ through a one-to-one correspondence between the Q gradients of $\Omega_{\mathcal{F}}$ and $\Omega_{\mathcal{R}}$. Specifically, for each perturbation, the projection is formulated as:

$$\nabla_{\mathcal{R}}^{\prime \xi_{\mathcal{F}}^{i}} = \nabla_{\mathcal{R}}^{\xi_{\mathcal{F}}^{i}} - \frac{\nabla_{\mathcal{R}}^{\xi_{\mathcal{F}}^{i}} \cdot \nabla_{\mathcal{F}}^{i}}{\|\nabla_{\mathcal{F}}^{i}\|_2^2} \cdot \nabla_{\mathcal{F}}^{i}. \tag{21}$$

where $i \in \{1, 2, \ldots, Q\}$. Through this iterative projection across all components, the updated gradient set $\Omega_{\mathcal{R}}^{\prime}$ for model $\mathcal{R}$ is formally derived as:

$$\Omega_{\mathcal{R}}^{\prime} = \left\{ \nabla_{\mathcal{R}}^{\prime \xi_{\mathcal{F}}^{1}}, \nabla_{\mathcal{R}}^{\prime \xi_{\mathcal{F}}^{2}}, \ldots, \nabla_{\mathcal{R}}^{\prime \xi_{\mathcal{F}}^{Q}} \right\}. \tag{22}$$

Finally, by incorporating these aligned and updated gradients into the generation process, Eq. (19) can be updated to yield the final adversarial example:

$$\xi_{\mathcal{R}}^{\prime} = \Pi_{\epsilon} \left\{ \xi_{\mathcal{F}}^{Q} - \mu \cdot \text{text} \left[\nu_2 \cdot \left(\nabla_{\mathcal{R}}^{\prime \xi_{\mathcal{F}}^{Q}} + \nu_3 \cdot \sum_{i=1}^{Q-1} \nabla_{\mathcal{R}}^{\prime \xi_{\mathcal{F}}^{i}} \right) \right] \right\}. \tag{23}$$

The process ensures that the gradient applied by each model has minimal interference with other models through simple calculation, and the problem of gradient conflict is alleviated. The overall process of the ICGA algorithm is shown in Algorithm 2.

Algorithm 2 Iterable cross-model gradient averaging adversarial sample generation.

Require:
Input: Attack image x, object detection surrogate models set $U = \{\mathcal{F}, \mathcal{R}\}$, number of outer iterations O, number of inner iterations Q.
Initialization: Adversarial perturbation ϵ, hyperparameters ν_1, ν_2, ν_3, learning rate μ.
Output: Adversarial sample x_{adv}.

$x_{adv} = x$
for o **in** $1, \ldots, O$ **do**
  Choose a model branch from U, for example $\mathcal{F}$.
  Update $\sigma_{\mathcal{F}} = \{\emptyset\}$.
  for i **in** $1, \ldots, Q$ **do**
    Calculate $L_{\mathcal{F}}$ according to Eq. (5).
    Calculate $\xi_{\mathcal{F}}^{i}$ and $\bigtriangledown_{\mathcal{F}}^{i}$ based on $L_{\mathcal{F}}$.
    Add $\xi_{\mathcal{F}}^{i}$ to $\sigma_{\mathcal{F}}$, $\bigtriangledown_{\mathcal{F}}^{i}$ to $\Omega_{\mathcal{F}}$.
    Update $x_{adv} = x_{adv} + \xi_{\mathcal{F}}^{i}$.
  Compute $\Omega_{\mathcal{R}}$ based on $\sigma_{\mathcal{F}}$ from another branch $\mathcal{R}$.
  end for
  for i **in** $1, \ldots, Q$ **do**
    Calculate $\bigtriangledown_{\mathcal{R}}^{\prime \xi_{\mathcal{F}}^{i}}$ according to Eq. (21).
    Add $\bigtriangledown_{\mathcal{R}}^{\prime \xi_{\mathcal{F}}^{i}}$ to $\Omega_{\mathcal{R}}^{\prime}$.
  end for
  Update $\xi_{\mathcal{R}}^{\prime}$ according to Eq. (23).
  Update $x_{adv} = x_{adv} + \xi_{\mathcal{R}}^{\prime}$.
end for
return x_{adv}

4. Experiments

4.1. Experimental setup

Datasets. The PASCAL VOC2007 (Lin et al., 2014) and MS COCO 2017 (Everingham et al., 2010) datasets are widely used benchmarks for object detection tasks. The COCO dataset includes 118,287 training images and 5000 validation images, covering 80 categories. The VOC dataset consists of 5011 training images and 4952 testing images, covering 20 categories. In our experiments, during the training phase, we used the COCO training set to train the surrogate model based on CTSRA. During the testing phase, we used the PASCAL and COCO datasets to evaluate the performance of the attacks. We randomly selected 500 images containing multiple (2–6) objects from the VOC testing set and the COCO validation set datasets for the attacks. When testing on the VOC dataset, we only consider objects present in the VOC.

Model. For a comprehensive and thorough analysis and comparison, we select several different architectures from the MMDetection codebase to evaluate our attack scheme, including two-stage detectors: Faster R-CNN (Ren et al., 2015), Libra R-CNN (Pang et al., 2019), and DCN (Dai et al., 2017); single-stage detector: RetinaNet (Lin et al., 2017),YOLO (Redmon & Farhadi, 2018); transformer-based detector: DETR (Carion et al., 2020) anchor-free detectors: FoveaBox (Kong et al., 2020), FreeAnchor (Zhang et al., 2019), and FCAF (Zhu et al., 2019). We used Faster R-CNN and Retina as our ensemble surrogate models. In addition, we selected PSPNet (Zhao et al., 2017) and DeepLabv3 (Chen et al., 2017) from the MMSegmentation codebase as auxiliary task models.

Competitors. We compare the directional adversarial attack performance of our proposed method with TOG (Chow et al., 2020), Context-aware Attack (CAA) (Cai et al., 2022), TSIAA (Sun et al., 2024a), and OFTA (Zhang et al., 2025).

Evaluation Metrics. We use Attack Success Rate (ASR) (Cai et al., 2022) to evaluate the performance of adversarial attacks on any victim object detector. Since the attacks are targeted misclassification attacks, we only focus on the victim object and define the ASR as the percentage of attacks where the victim object is successfully misclassified into the target label. During the experiment, we consider an attack successful if the target object is detected with an Intersection over Union (IoU) greater than 0.3; otherwise, the attack is considered a failure. To select the target object, we randomly choose a label that does not exist in the original image, ensuring that only new misclassifications are counted as successful attacks.

Implementation Details. Our method is implemented on PyTorch and utilizes one NVIDIA RTX 3090 GPU for all experiments. For the CTSRA strategy, we set the batch size to 8, the maximum number of iterations to 12, and used stochastic gradient descent with a momentum of 0.9 for training. The weight decay is set to 0.0001, and the learning rate is 0.01. We set the hyperparameter α to 200. We resized all images to 1333×800 and applied data augmentation with a 50% probability of flipping. For the ICGA strategy, we set the perturbation to $L_\infty \leq \{10, 20, 30\}$; the step size μ is set to 1.5, the number of external iterations O is set to 200 to ensure attack effectiveness, and the number of internal iterations Q is set to 5. Additionally, we initialize (ν_1, ν_2, ν_3) to $(0.5, 0.5, 0.01)$.

4.2. Experimental results

In Table 1, we report the ASR results of our method on the VOC and COCO datasets under different perturbation ranges, and compare them with other state-of-the-art targeted attack methods, we selected FRCNN and YOLOv3 as the white-box models, as they achieved the best attack performance in their original papers. During the attack process, we selected two white-box surrogate models, Faster R-CNN and RetinaNet, which are identical to those used in our proposed method. Specifically, during each iteration of adversarial example generation, we obtained

Table 1
For the VOC and COCO datasets, we evaluate the ASRs of white-box and black-box attacks with different perturbation budgets ($L_\infty \leq \{10, 20, 30\}$). A higher ASR value indicates better attack performance. We present a comparison with the TOG and CAA methods. The abbreviations are as follows: Faster R-CNN (FRCNN), RetinaNet (Retina), Libra R-CNN (Libra), FoveaBox (Fovea), FreeAnchor (Free), DCN, FSFA, YOLO, DETR.

Datasets		COCO									VOC								
Perturbation	Methods	WhiteBox		BlackBox							WhiteBox		BlackBox						
		w1	w2	b1	b2	b3	b4	b5	b6	b7	w1	w2	b1	b2	b3	b4	b5	b6	b7
$L_\infty \leq 10$	TOG	29.0	32.2	4.8	11.6	16.6	-	-	-	-	40.0	53.8	9.2	22.2	27.4	–	–	–	–
	CAA	41.2	54.4	11.2	18.6	25.0	16.8	40.0	–	10.8	55.8	75.6	20.4	33.6	39.2	27.8	50.6	–	20.2
	TSIAA	52.0	52.8	24.6	31.0	31.6	26.4	29.6	6.6	9.8	56.6	57.0	30.2	36.2	36.2	31.4	34.0	12.0	18.0
	OFTA	46.0	47.0	18.8	25.0	26.2	20.4	23.6	10.8	12.2	52.6	53.2	24.6	30.0	31.2	26.0	29.0	18.2	21.8
	EVERY	38.2	44.0	8.8	16.6	22.2	18.0	22.0	8.8	12.0	48.8	50.2	14.4	26.0	30.4	24.8	25.0	14.8	18.6
	CTSTA	76.8	69.8	33.4	37.4	48.4	60.6	51.2	15.0	19.8	74.4	70.8	30.2	37.6	46.6	56.0	59.8	20.4	27.2
$L_\infty \leq 20$	TOG	51.8	49.2	11.8	22.0	28.6	–	–	–	–	65.2	67.8	21.4	34.4	41.8	–	–	–	–
	CAA	64.4	70.0	22.2	35.4	40.8	30.4	60.8	–	20.0	78.6	87.2	38.4	51.6	56.6	44.6	76.0	–	34.0
	TSIAA	62.6	63.0	30.0	35.8	37.0	31.2	34.4	10.0	12.2	67.6	69.0	36.0	41.2	42.8	25.8	39.2	16.4	20.0
	OFTA	57.0	56.6	24.0	29.4	31.4	26.0	27.0	16.8	18.0	62.0	62.6	30.4	35.8	37.8	32.4	33.6	21.2	27.8
	EVERY	55.8	54.2	20.4	26.4	30.8	24.0	26.2	12.8	16.4	60.0	62.2	28.6	34.0	35.2	30.6	30.0	18.4	26.0
	CTSTA	80.8	75.8	37.4	41.4	55.6	65.8	63.8	18.2	21.8	89.0	80.2	39.4	47.8	60.8	65.2	78.8	25.8	35.2
$L_\infty \leq 30$	TOG	57.6	54.4	15.4	25.6	34.8	–	–	–	–	70.6	70.4	28.6	41.6	48.0	–	–	–	–
	CAA	68.6	75.4	27.2	39.2	46.2	31.4	66.0	–	21.2	80.6	88.0	44.2	56.8	63.6	50.4	79.2	–	40.2
	TSIAA	68.2	68.4	35.8	40.8	42.0	36.8	38.8	12.4	14.8	73.0	74.2	41.8	46.0	48.0	41.8	43.8	18.8	22.8
	OFTA	61.6	61.0	29.6	34.4	35.6	31.0	32.2	18.0	21.0	67.8	67.8	36.4	41.2	43.6	37.6	38.8	24.0	29.0
	EVERY	60.0	59.2	22.4	28.0	31.4	30.0	29.8	16.8	20.0	65.0	64.2	31.4	37.8	40.0	32.8	35.0	20.4	27.8
	CTSTA	83.8	76.8	39.4	51.6	63.6	72.8	69.8	20.0	27.6	90.2	84.2	47.8	59.2	67.4	78.0	80.4	28.8	42.0

the gradients of the target attack task from both models separately. We then calculated the arithmetic mean of these two gradients to serve as the final gradient update direction. In our experiments, we also selected FRCNN and Retina as the white-box surrogate models for object detection, and adopted PSPNet and DeepLabv3 as the teacher models in the CTSRA process. For convenience, in the table, the two white-box models are denoted as w1 and w2, and the five black-box models are denoted as b1, b2, b3, b4, b5, b6, and b7, respectively.

Under the white-box targeted attack setting, CTSTA demonstrates competitive attack performance in terms of ASR compared to existing methods. Even in white-box scenarios, the ASR may still be affected due to the inherent challenge of targeted misclassification attacks. Nevertheless, our method shows stable improvements on this task. Across different perturbation levels on both the VOC and COCO datasets, CTSTA exhibits overall strong performance and surpasses existing methods in most cases.

In the targeted attack black-box setting, CTSTA also demonstrates superior transferability over other approaches. It can be observed that, although the success rate of black-box attacks is significantly lower than that of white-box attacks, the proposed attack strategy still provides noticeably better attack transfer success rates compared to the comparison methods. For the VOC and COCO datasets, compared to the baseline methods, our approach consistently achieves higher success rates, with improvements reaching up to 36.0%. This represents a significant improvement for the well-known challenge of transfer attacks on object detectors in black-box settings. Furthermore, on detectors with significantly different architectures such as YOLO and DETR, although the performance of our method is relatively weaker, it still significantly outperforms the comparison methods, which precisely demonstrates its core advantage of reducing dependency on architectural similarity and maintaining cross-architecture transferability.

A visual comparison result display taking the CAA method as an example. We present the detection results of multiple attack examples on the black-box model Libra in Figs. 3 and 4, respectively. The first column indicates the victim category and the target category of the attack in each example, the second column shows the detection results on the clean image, the third column displays the detection results on the adversarial examples generated by the each comparison method, and the fourth column shows the detection results on the adversarial examples generated by our method. From the visualization results, we can notice the following observations: a) In the three examples in Fig. 3, the adversarial samples generated by the CAA method sometimes still detect the victim object or fail to detect any objects around the victim object, resulting in an unsuccessful attack. However, the adversarial samples generated by our method successfully achieve the attack in all cases; b) In the three examples in Fig. 4, both the CAA method and our method successfully achieve the attack with the generated adversarial samples. However, it can be seen that our method achieves higher confidence in detecting the target object; c) Additionally, the CAA method exploits co-occurrence between objects to perform the attack, leading to cluttered detection results with unrelated objects appearing. In contrast, our method achieves more precise attacks that only target the specified object.

4.3. Ablation study

FRCNN and Retina achieved the highest average ASR when used as white-box surrogate models. To select appropriate surrogate models, we randomly constructed multiple combinations of white-box and black-box detectors from several mainstream object detection models, and evaluated the ASR of each combination on the black-box models. As shown in Table 2, clear differences in transferability were observed across combinations, which fundamentally stem from the degree of gradient space alignment. Although FRCNN and Retina differ in architecture, they form gradient complementarity: FRCNN's region proposal mechanism is sensitive to positional information, while RetinaNet's Focal Loss is robust to class imbalance. Together, they cover a broader range of gradient directions, enabling strong transferability across diverse black-box architectures. In contrast, combinations involving FoveaBox or Libra R-CNN yield suboptimal transferability due to poor gradient alignment or weak complementarity. In summary, the FRCNN and Retina combination achieves the best average ASR, demonstrating strong generalization capability.

The order of attack model selection does not affect the final performance of generating adversarial examples. As shown in Table 3, we

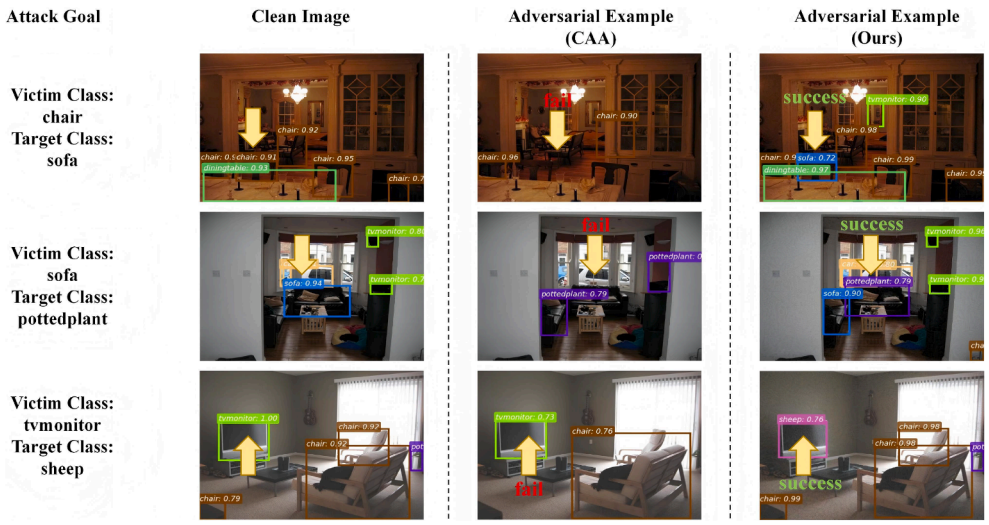


Fig. 3. Adversarial examples where the CAA attack fails, but our CTSRA attack succeeds. For example, in the third line example, the goal is to attack the TV monitor as a sheep, our method achieves a successful attack, but the CAA method still detects the TV monitor.

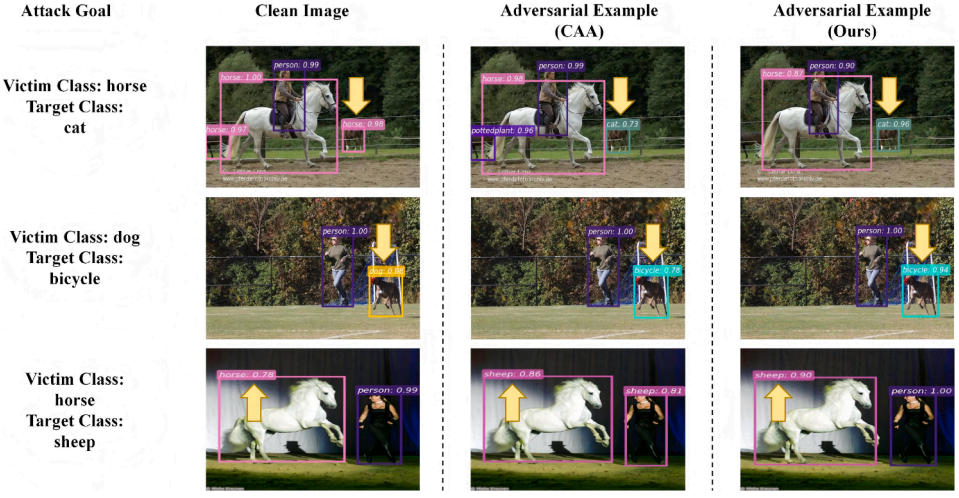


Fig. 4. Adversarial examples where both the CAA and our CTSRA attack succeed, but our method achieves higher target object detection confidence. For example, in the first line example, the purpose is to attack a horse as a cat, and our method achieves a successful attack with a confidence of 0.96, while the CAA method also achieves a successful attack with a confidence of only 0.73.

Table 2
Comparison of ASR performance under different combinations of white-box surrogate models.

Models	b1	b2	b3	b4	b5
FRCNN + Libra	-	30.2	39.4	59.6	42.4
FRCNN + Fovea	26.2	-	47.4	51.6	52.4
Retina + Libra	-	16.2	27.6	37.0	30.0
Retina + Fovea	14.2	-	22.0	23.4	31.6
FRCNN + Retina	33.4	37.4	48.6	60.6	51.2

Table 3
ASRs on the VOC dataset with different surrogate model selection orders.

Methods	Whitebox		Blackbox				
	w1	w2	b1	b2	b3	b4	b5
$\mathcal{F} \rightarrow \mathcal{R}$	76.8	69.8	33.4	37.4	48.4	60.6	51.2
$\mathcal{R} \rightarrow \mathcal{F}$	78.0	69.0	35.0	36.8	49.0	58.8	50.0

evaluate the impact of different attack sequences on the ASRs. $\mathcal{F} \rightarrow \mathcal{R}$ indicates that model FRCNN is selected first and then model Retina is selected, while $\mathcal{R} \rightarrow \mathcal{F}$ indicates that model Retina is selected first and then model FRCNN is selected. It can be observed that the attack results are almost identical, and considering the result fluctuations caused by randomly selecting attack targets during the attack process, it can be concluded that the model selection order does not affect the attack performance.

To better verify the efficacy of each component of our CTSRA attack method, we conduct extensive ablation studies on the following variants: To better verify the effectiveness of each component in our proposed CTSRA attack method, we conduct extensive ablation studies on the following variants: (1) We use an object detection surrogate model and directly apply weighted averaging of adversarial losses from different models as the baseline; (2) 'CTSRA': Based on the baseline, we add the CTSRA module to enhance the expressive capability of the surrogate model; (3) 'ICGA': We further introduce the ICGA module to alleviate gradient conflicts. As shown in Table 2, we selected the three most beneficial white-box model combinations from Table 4 to evaluate the effectiveness and generalization of each component in our method. The

Table 4
Ablation study on the effectiveness and generalization of each CTSTA component under different white-box model combinations.

Models	Methods		Blackbox						
	CTSRA	ICGA	b1	b2	b3	b4	b5	b6	b7
FRCNN + Libra			–	20.4	30.0	49.2	33.6	3.0	4.6
	✓		–	23.2	33.4	56.6	40.0	6.2	7.4
		✓	–	22.4	38.0	52.4	40.0	8.0	9.8
	✓	✓	–	30.2	39.4	59.6	42.4	12.4	13.8
FRCNN + Fovea			17.0	–	34.0	46.0	42.0	4.8	5.2
	✓		24.0	–	43.4	49.6	44.4	7.4	9.6
		✓	23.0	–	43.2	49.0	43.6	10.8	13.0
	✓	✓	26.2	–	47.4	51.6	52.4	14.2	17.6
FRCNN + Retina			17.0	27.8	40.0	47.4	43.0	6.6	7.4
	✓		24.4	31.4	44.4	56.0	47.6	9.8	11.4
		✓	28.8	33.6	48.0	54.2	47.0	12.6	17.8
	✓	✓	33.4	37.4	48.6	60.6	51.2	15.0	19.8

Table 5
Comparison of ASR between CTSTA and baseline methods under common adversarial defenses.

Metric	Methods	Black Models				
		b1	b2	b3	b4	b5
CON	TSIAA	0.74	0.61	0.79	0.65	0.53
	OFTA	0.73	0.56	0.69	0.51	0.60
	CTSTA	0.81	0.62	0.86	0.69	0.64
Non-mAP	TSIAA	0.82	0.78	0.81	0.79	0.80
	OFTA	0.77	0.75	0.81	0.76	0.78
	CTSTA	0.86	0.87	0.86	0.85	0.89

presence or absence of checkmarks indicates whether the corresponding module is applied.

Each component of our CTSTA method plays a crucial role in enhancing the ASR. As shown in Table 4, by comparing the ASR of different variants on black-box and white-box models in the COCO dataset, we draw the following conclusions: (1) Across all model groups, comparing the results of the Baseline and the variant with CTSRA, the ASR of black-box models increases by approximately 2.4% to 9.4% on average. This result strongly supports the effectiveness of leveraging semantic segmentation auxiliary tasks through CTSRA to enrich and optimize object detection tasks. (2) Comparing the Baseline with the variant incorporating ICGA, the ASR improves by approximately 1.6% to 11.8% on average, indicating that the ICGA module effectively captures inter-model correlations, mitigates gradient conflicts, reduces oscillations, and improves the effectiveness of adversarial attacks. (3) When both CTSRA and ICGA are used together, the ASR of each black-box model is further improved compared to using either module alone. This demonstrates that the components of our method complement each other during the attack process, helping the model capture more generalized features, stabilize gradient updates, and significantly enhance attack transferability. (4) Comparing the results across different white-box surrogate model combinations, both CTSRA and ICGA contribute to ASR improvements, further validating the generality and effectiveness of our proposed method. (5) Through ablation comparisons with and without the CTSRA component, we verify the role of semantic segmentation supervision in alleviating architecture-induced attack overfitting. Experimental results show that, compared to variants without CTSRA, introducing the semantic segmentation auxiliary task significantly improves attack transferability across detectors with different architectures, such as DETR and YOLO. This indicates that CTSRA helps reduce reliance on specific detection structures, thereby enhancing the generalization ability of cross-architecture attacks.

Table 6
Comparison of different auxiliary tasks for adversarial attack performance.

Auxiliary Task	b1	b2	b3	b4	b5
Depth Estimation	21.4	28.6	34.8	41.2	45.0
Edge Detection	17.6	23.4	29.0	33.8	36.2
Semantic Segmentation	30.2	37.6	46.6	56.0	59.8

Table 7
Comparison of information sufficiency before and after semantic segmentation enhancement..

Model	b1	b2	b3	b4	b5
Retina	0.0723	0.0346	0.0576	0.1047	0.1206
Retina + DeepFool	0.4219	0.4099	0.3703	0.3381	0.2701

Analysis of Attack Quality and Impact on Non-target Objects. To more comprehensively evaluate attack performance, we introduce two metrics: the average confidence of the target category after successful attacks (CON) and the Non-target Object mAP (Non-mAP). CON measures the confidence of the target category predictions after an attack. As shown in Table 5, CTSRA adversarial samples not only successfully mislead the model but also achieve significantly higher confidence in the target category compared to other methods, indicating that the attack is more persuasive and robust. Non-mAP measures the side effects on non-target objects, using the detection results of clean samples as the baseline. Results show that CTSRA maintains a high level of Non-mAP while significantly outperforming baseline methods, indicating that the perturbation energy is primarily focused on the attacked target, minimizing interference with the background and unrelated objects.

Auxiliary Task Comparison. We additionally select depth estimation (mono + stereo) and edge detection (BSDS500) as auxiliary tasks. As shown in Table 6, under the same experimental settings, attacks guided by semantic segmentation achieve superior transferability and overall attack performance compared to those using other auxiliary tasks. This indicates that the pixel-level dense priors provided by semantic segmentation can more effectively compensate for the global semantic information missing due to sparse annotations in object detection, thereby enabling perturbations to better exploit shared semantic vulnerabilities across different models.

Validation of the Effectiveness of Adversarial Information Supplementation. Let f_s denote the surrogate model, f_t the target model, x the input, y the label, and L the loss function, the gradients: $g_s(x) = \nabla_x L(f_s(x), y), g_t(x) = \nabla_x L(f_t(x), y)$. We define the degree of information insufficiency as the average directional deviation between the surrogate and target gradients over the data distribution D: $\mathcal{I}(f_s, f_t) = \mathbb{E}_{x\sim D}\left[\cos\langle g_s(x), g_t(x)\rangle\right]$, the larger $\mathcal{I}$ indicates that the adversarial information provided by the surrogate model f_s is fully consistent with that of the target model f_t. To validate the effectiveness of adversarial information supplementation, we adopt Retina as the baseline surrogate model and compare its perfrmance before and after semantic segmentation enhancement. As shown in Table 7, the enhanced Faster R-CNN achieves higher $\mathcal{I}$ values across all five target models, including Libra R-CNN, FoveaBox, FreeAnchor, DCN, and FSAF, demonstrating that it provides stronger adversarial information.

Analysis of hyper-parameters selection. We further analyze four key hyper-parameters in our CTSRA attack, namely the scaling factor α that controls the influence of semantic consistency weights in the CTSRA loss, the number of historical iterations Q used in the ICGA module, the weight of historical gradients v_3 in the ICGA module, and the number of external iterations O used in the HGS module. In our experiments, each hyper-parameter setting was tested multiple times, and the average results were reported to ensure the stability and reliability of the data. (1) As shown in Fig. 5 (a), for the hyper-parameter α, which controls the im-

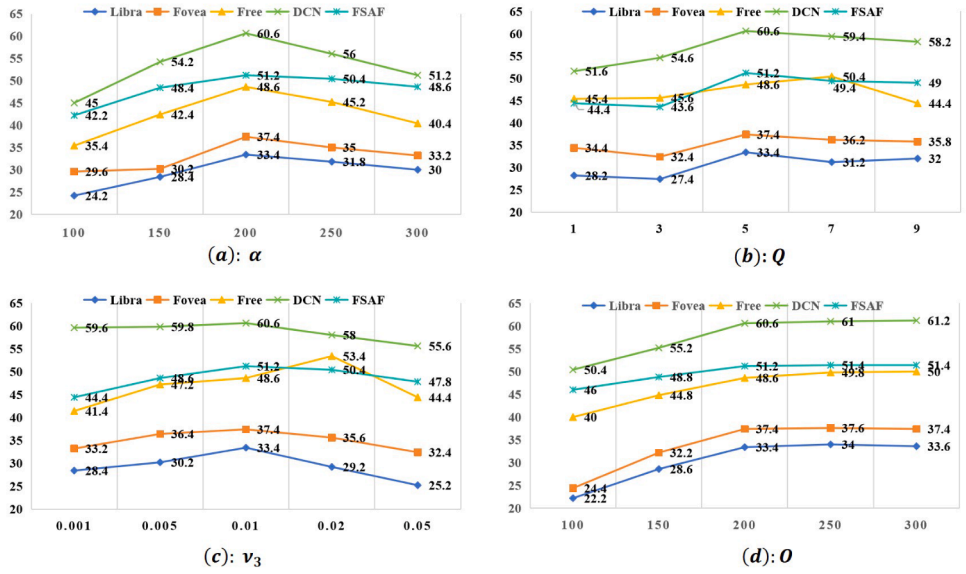


Fig. 5. ASRs under different hyper-parameter settings in CTSRA: α (semantic weight scale), Q (historical iteration count), v_3 (historical gradient weight), and O (external iterations of HGS).

pact strength of the semantic consistency weight $\eta_{m,n}$ in the CTSRA loss, we conducted an ablation study with values $\alpha \in 100, 150, 200, 250, 300$. The ASR gradually increases when α increases from 100 to 200, indicating that the guidance from semantic consistency is effectively utilized. However, when α continues to increase, the performance slightly drops, possibly due to the overemphasis on semantic consistency, which may suppress some beneficial but less consistent features. Considering both performance and robustness, we finally set α to 200. (2) As shown in Fig. 5 (b), we test the attack performance of multiple black-box models on different datasets under different internal iteration numbers Q set to $\{1, 3, 5, 7, 9\}$. When Q increases from 1 to 5, the ASRs significantly improve with the increase in Q. However, as Q continues to increase, the improvement in attack performance gradually plateaus. Since increasing Q also increases the computational cost, we set Q to 5 to achieve the best balance between attack performance and efficiency. (3) As shown in Fig. 5 (c), we test the attack results on various black-box models on the VOC dataset under different v_3 values. When the v_3 value increases from 0.001 to 0.01, the ASRs gradually increases due to the integration of historical gradients. However, as the v_3 value continue to increase, the ASRs decrease because the current gradient update direction is neglected. Therefore, setting the v_3 value to 0.01 achieve the optimal ASRs. (4) As shown in Fig. 5 (d), To investigate the impact of the external iteration number O on the effectiveness of our CTSTA method, we perform a comprehensive analysis by varying O across the values 50, 100, 150, 200, 250. We observe a consistent improvement in the Attack Success Rate (ASR) as O increases from 50 to 200, highlighting the benefit of more extensive optimization in capturing robust adversarial directions. This trend demonstrates that increasing O allows the model to explore a broader gradient landscape and generate more effective perturbations. However, when O exceeds 200, the improvement in ASR becomes marginal while the computational overhead increases significantly. Therefore, to ensure an efficient trade-off between attack effectiveness and computational cost, we select $O = 200$ as the optimal setting in all subsequent experiments.

Computational Overhead Analysis. CTSRA incurs most of its computational cost during the offline distillation pre-training phase, which enhances the feature representation of the surrogate model through a semantic segmentation auxiliary task. Once pre-training is completed, the online attack phase no longer requires the teacher model or distillation loss computation, and the generation speed of adversarial examples is on the same order of magnitude as the baseline methods. As shown in Table 8, although the time to generate a single sample is slightly higher than some baseline methods, it still achieves significant advantages in the key metric ASR.

Table 8
Comparison of average inference time for different adversarial attack methods.

Methods	CAA	TSIAA	OFTA	CTSTA
time/s	25	40	25	30

Table 9
Comparison of ASR between CTSTA and baseline methods under common adversarial defenses.

Defense	Methods	Black Models				
		Libra	Fovea	Free	DCN	FSAF
Random Resize	CAA	7.8	12.6	19.0	9.8	35.4
	TSIAA	17.6	22.2	23.2	18.6	20.6
	OFTA	13.2	17.6	19.2	13.4	16.4
	CTSTA	21.2	24.4	30.2	39.4	47.6
AT	CAA	8.2	16.8	23.2	11.0	38.6
	TSIAA	19.6	23.8	26.2	21.4	23.6
	OFTA	15.6	20.2	21.8	16.0	19.2
	CTSTA	26.2	29.0	39.6	48.0	43.4
NRP	CAA	8.2	14.0	21.0	10.4	37.0
	TSIAA	18.2	23.0	24.4	20.0	22.0
	OFTA	14.2	18.8	20.0	14.8	17.8
	CTSTA	23.8	26.6	34.8	43.4	40.0

Robustness Evaluation Under Common Adversarial Defenses. To assess the robustness and practical applicability of the proposed CTSTA method in realistic, defense-aware scenarios, we conducted experiments under several widely-used adversarial defense techniques, including input transformation defenses, Neural Representation Purification and adversarial training. For input transformation defenses, we use Random Resize: each image is randomly scaled with a scale factor sampled from

Table 10
Comparison of attack performance with different numbers of aggregated surrogate models.

Models	b1	b2	b3	b4	b5	Time (s)
FRCNN + Retina	33.4	37.4	48.6	60.6	51.2	25
FRCNN + Retina + Libra	37.2	41.8	53.4	64.0	55.6	38
FRCNN + Retina + Fovea	37.4	42.2	54.0	65.2	56.8	38

the range [0.5,3.0], and then resized back to the original dimensions via bilinear interpolation. Additionally, we employed adversarial training, where adversarial samples generated by FGSM (with an l_∞ perturbation bound of 10) are incorporated into the training set to improve model robustness. As shown in Table 9, CTSTA maintains strong performance under all the above defense strategies. Under both input transformation and adversarial training conditions, although the method exhibits some decline in transferability and attack success rate, it still achieves the best performance among all comparative methods, demonstrating its robustness and practical effectiveness in real-world scenarios.

Analysis of the relationship between the number of aggregated models and attack performance. We further experiment by adding more models to the candidate pool beyond the two-model aggregation. As shown in Table 10, three-model aggregation further improves the attack success rate compared to two-model aggregation, benefiting from gradient complementarity across more models that cover a broader gradient space, though computational cost also increases accordingly. Since most mainstream baseline methods adopt two-model ensemble attacks, we use two-model aggregation in our main experiments to maintain consistency with existing work. In practice, the number of aggregated models can be flexibly selected based on the desired attack effectiveness and time budget.

5. Conclusion

In this study, we approach the problem from a multi-task learning perspective and propose an innovative cross-task synergistic transferable attack. By leveraging a semantic segmentation auxiliary task, which is highly relevant to object detection, we effectively address the deficiencies in the adversarial information of the surrogate model, significantly enhancing attack transferability. We conducted extensive experiments on the COCO and VOC large-scale datasets, and the results demonstrate that our attack achieves a significantly higher success rate compared to the baseline methods, validating the effectiveness of our approach. Additionally, ablation studies confirm the effectiveness of each component of our method. However, the proposed method still has certain limitations. In the preparation phase of the white-box surrogate models, we employ Cross-Task Semantic Representation Alignment to enable the surrogate models to better capture key features and attack-relevant information in object detection tasks. This process requires considerable training time. In practical scenarios, to achieve more robust attack performance, it is often necessary to conduct distillation training and hyperparameter tuning for multiple surrogate models individually. Each model's training involves significant computational resources and time, leading to increased overall computational cost. As a result, this stage becomes a bottleneck that needs to be carefully addressed and improved for the method to be widely adopted in real-world or large-scale applications.

Social Impacts. The object detection models are widely used in real-world applications, making their security critically important. Our research is not intended for malicious use of these technologies but rather to explore the vulnerabilities of these models when faced with various attacks. This work aims to provide valuable insights for future model design and defense strategies. In future work, we plan to continue our research into relevant defense technologies and explore new solutions to address the evolving security challenges.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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